

EMPLOYEE ATTRITION ANALYSIS

Project Report:

Submitted By

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Submitted to

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Objective:

The primary objective of this project is to analyze employee attrition using data analytics techniques and identify the key factors influencing employee turnover. The project aims to clean and preprocess HR data using Python, store the processed data in a MySQL database, and develop interactive Power BI dashboards to visualize employee attrition trends. Through exploratory data analysis, KPI calculations, and business intelligence reporting, the project provides actionable insights and recommendations that can help organizations improve employee retention and support data-driven HR decision-making.

Internship Details

Project Title	HR Attrition Analysis
Tools Used	Python, Pandas, Numpy, SQL, Power BI
Dataset	HR Employee Attrition

Dataset Understanding in Python:

1. Import Libraries

```
Employee_Attrition.ipynb X
C:\Users\ocdga\Desktop\Internship Project\Project Report> Employee_Attrition.ipynb > df.shape
Generate + Code + Markdown | Run All Restart Clear All Outputs | Jupyter Variables Outline ... Python 3.14.0

# Import Libraries

import pandas as pd
import numpy as np

[1] ✓ 43s Python
```

2. Load dataset

```
# load the dataset
df = pd.read_csv(r"C:\Users\ocdga\Desktop\Internship Project\Project Report\HR-Employee-Attrition Cleaned Dataset.csv")

[13] ✓ 0.0s Python
```

3. Rows and Columns

```
# Rows and Columns
df.shape

[25] ✓ 0.0s Python

... (1470, 35)
```

4. Columns Names

```
# Column Names
df.columns

[36] ✓ 0.0s Python

... Index(['Age', 'Attrition', 'BusinessTravel', 'DailyRate', 'Department',
'DistanceFromHome', 'Education', 'EducationField', 'EmployeeCount',
'EmployeeNumber', 'EnvironmentSatisfaction', 'Gender', 'HourlyRate',
'JobInvolvement', 'JobLevel', 'JobRole', 'JobSatisfaction',
'MaritalStatus', 'MonthlyIncome', 'MonthlyRate', 'NumCompaniesWorked',
'OverTime', 'PercentSalaryHike', 'PerformanceRating',
'RelationshipSatisfaction', 'StandardHours', 'StockOptionLevel',
'TotalWorkingYears', 'TrainingTimesLastYear', 'WorkLifeBalance',
'YearsAtCompany', 'YearsInCurrentRole', 'YearsSinceLastPromotion',
'YearsWithCurrManager'],
dtype='str')
```

5. Dataset Information

```
# Dataset Information
df.info()
```

```
[27] ✓ 0.0s Python
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 35 columns):
 #   Column              Non-Null Count  Dtype
---  -
 0   Age                 1470 non-null  int64
 1   Attrition           1470 non-null  str
 2   BusinessTravel      1470 non-null  str
 3   DailyRate           1470 non-null  int64
 4   Department          1470 non-null  str
 5   DistanceFromHome    1470 non-null  int64
 6   Education           1470 non-null  int64
 7   EducationField       1470 non-null  str
 8   EmployeeCount       1470 non-null  int64
 9   EmployeeNumber      1470 non-null  int64
10   EnvironmentSatisfaction 1470 non-null  int64
11   Gender              1470 non-null  str
12   HourlyRate          1470 non-null  int64
13   JobInvolvement       1470 non-null  int64
14   JobLevel            1470 non-null  int64
15   JobRole              1470 non-null  str
16   JobSatisfaction      1470 non-null  int64
17   MaritalStatus        1470 non-null  str
```

6. Display First rows

```
# Display first rows
df.head()
```

```
[26] ✓ 0.0s Python
```

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EmployeeCount	EmployeeNumber	...	RelationshipS
0	41	Yes	Travel_Rarely	1102	Sales	1	2	Life Sciences	1	1	...	
1	49	No	Travel_Frequently	279	Research & Development	8	1	Life Sciences	1	2	...	
2	37	Yes	Travel_Rarely	1373	Research & Development	2	2	Other	1	4	...	
3	33	No	Travel_Frequently	1392	Research & Development	3	4	Life Sciences	1	5	...	
4	27	No	Travel_Rarely	591	Research & Development	2	1	Medical	1	7	...	

5 rows x 35 columns

```
# Dataset Information
df.info()
```

```
[27] ✓ 0.0s Python
```

7. Statistical Summary

```
# Statistical Summary
df.describe()
```

```
[28] ✓ 0.0s Python
```

	Age	DailyRate	DistanceFromHome	Education	EmployeeCount	EmployeeNumber	EnvironmentSatisfaction	HourlyRate	JobInvolvement	J
count	1470.000000	1470.000000	1470.000000	1470.000000	1470.0	1470.000000	1470.000000	1470.000000	1470.000000	1470.000000
mean	36.923810	802.485714	9.192517	2.912925	1.0	1024.865306	2.721769	65.891156	2.729932	2.729932
std	9.135373	403.509100	8.106864	1.024165	0.0	602.024335	1.093082	20.329428	0.711561	0.711561
min	18.000000	102.000000	1.000000	1.000000	1.0	1.000000	1.000000	30.000000	1.000000	1.000000
25%	30.000000	465.000000	2.000000	2.000000	1.0	491.250000	2.000000	48.000000	2.000000	2.000000
50%	36.000000	802.000000	7.000000	3.000000	1.0	1020.500000	3.000000	66.000000	3.000000	3.000000
75%	43.000000	1157.000000	14.000000	4.000000	1.0	1555.750000	4.000000	83.750000	3.000000	3.000000
max	60.000000	1499.000000	29.000000	5.000000	1.0	2068.000000	4.000000	100.000000	4.000000	4.000000

8 rows x 26 columns

Data Cleaning in Python:

1. Findout Missing values

```
# Findout Missing Values
df.isnull().sum()
```

[29] ✓ 0.0s Python

Age	0
Attrition	0
BusinessTravel	0
DailyRate	0
Department	0
DistanceFromHome	0
Education	0
EducationField	0
EmployeeCount	0
EmployeeNumber	0
EnvironmentSatisfaction	0
Gender	0
HourlyRate	0
JobInvolvement	0

2. Findout Duplicate values

```
# Findout Duplicate Values
df.duplicated().sum()
```

[30] ✓ 0.0s Python

np.int64(0)

3. Findout unique values

```
# Findout Unique Values
df.nunique()
```

[31] ✓ 0.0s Python

If the number of unique values is 1, it means the column is constant.

Age	43
Attrition	2
BusinessTravel	3
DailyRate	886
Department	3
DistanceFromHome	29
Education	5
EducationField	6
EmployeeCount	1
EmployeeNumber	1470
EnvironmentSatisfaction	4

4. Removing unnecessary columns

```
# Removing unnecessary columns
df = df.drop("Over18", axis=1)
```

[32] ✓ 0.0s Python

5. Column names

```
df.columns

Index(['Age', 'Attrition', 'BusinessTravel', 'DailyRate', 'Department',
       'DistanceFromHome', 'Education', 'EducationField', 'EmployeeCount',
       'EmployeeNumber', 'EnvironmentSatisfaction', 'Gender', 'HourlyRate',
       'JobInvolvement', 'JobLevel', 'JobRole', 'JobSatisfaction',
       'MaritalStatus', 'MonthlyIncome', 'MonthlyRate', 'NumCompaniesWorked',
       'OverTime', 'PercentSalaryHike', 'PerformanceRating',
       'RelationshipSatisfaction', 'StandardHours', 'StockOptionLevel',
       'TotalWorkingYears', 'TrainingTimesLastYear', 'WorkLifeBalance',
       'YearsAtCompany', 'YearsInCurrentRole', 'YearsSinceLastPromotion',
       'YearsWithCurrManager'],
      dtype='str')
```

6. Data frame Information

```
df.info()

<class 'pandas.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 34 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Age                    1470 non-null  int64
1   Attrition              1470 non-null  str
2   BusinessTravel         1470 non-null  str
3   DailyRate              1470 non-null  int64
4   Department             1470 non-null  str
5   DistanceFromHome       1470 non-null  int64
```

7. Dataframe Rows and Columns

```
df.shape

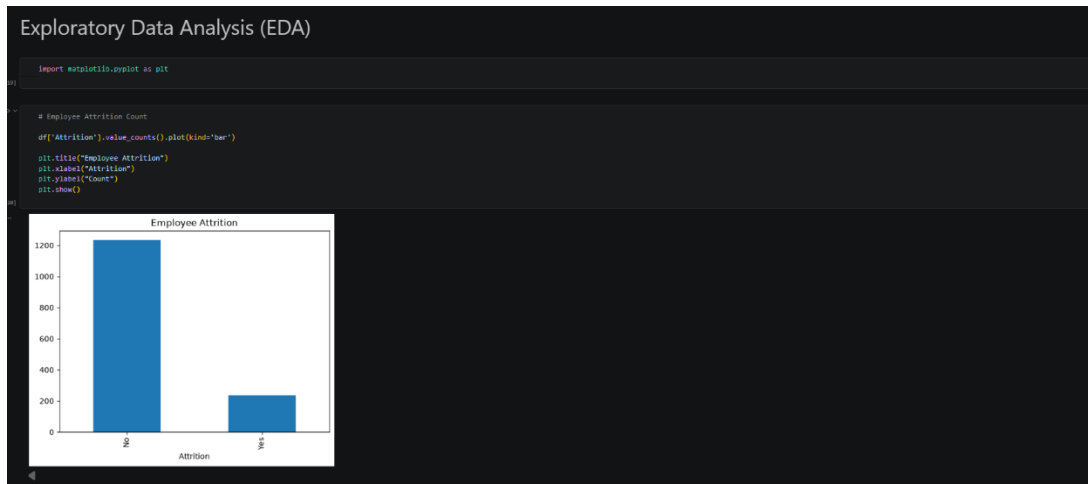
(1470, 34)
```

During data cleaning, the columns **EmployeeCount**, **Over18**, and **StandardHours** were identified as constant columns because each contained only a single unique value across all records. Since these columns did not provide any variability or analytical value, they were removed from the dataset to simplify the analysis and improve data quality.

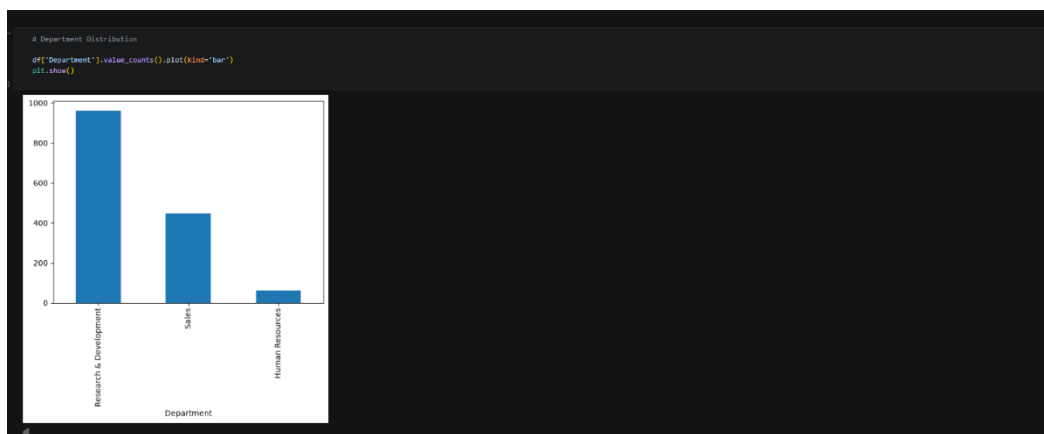
1. No missing values found
2. No duplicate records found
3. Dataset is clean and ready for analysis

Exploratory Data Analysis (EDA):

1. Employee Attrition Count



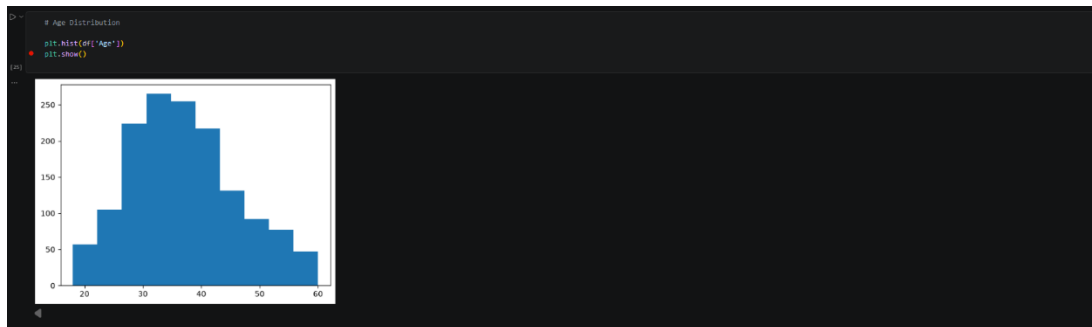
2. Department Distribution



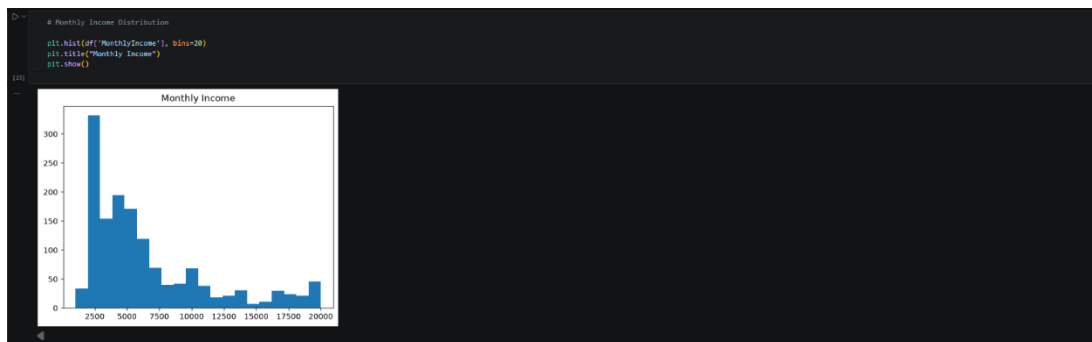
3. Gender Distribution



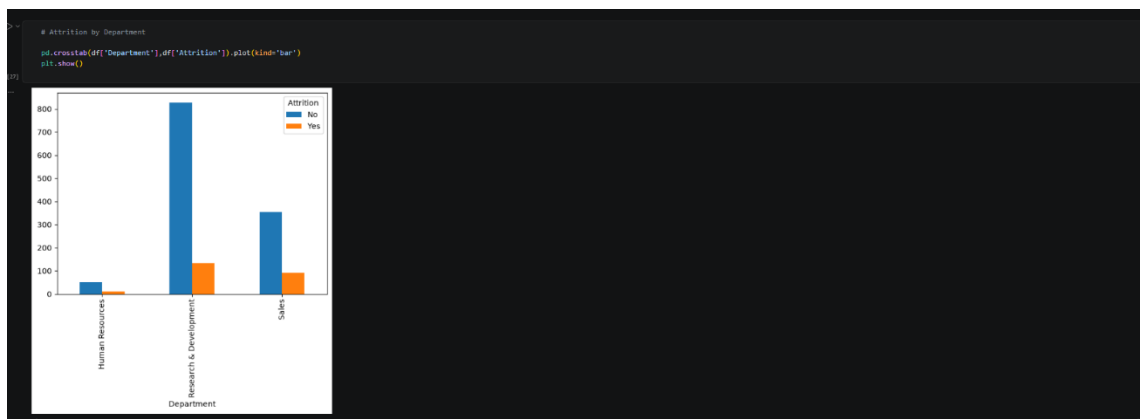
4. Age Distribution



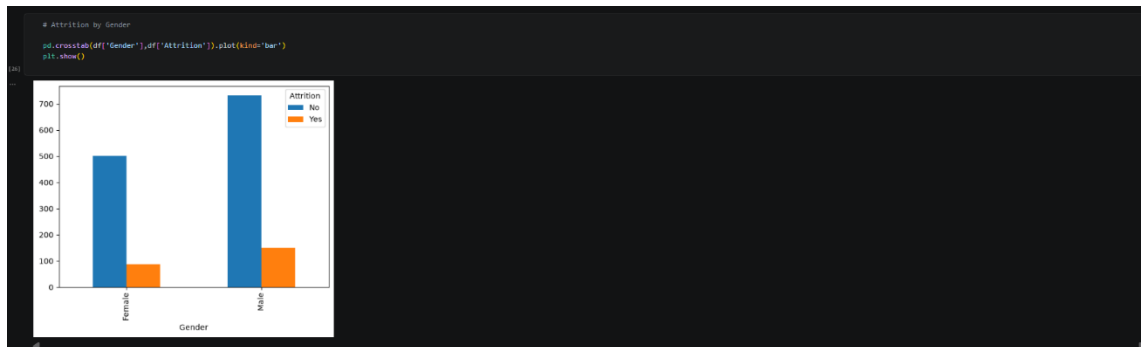
5. Monthly Income Distribution



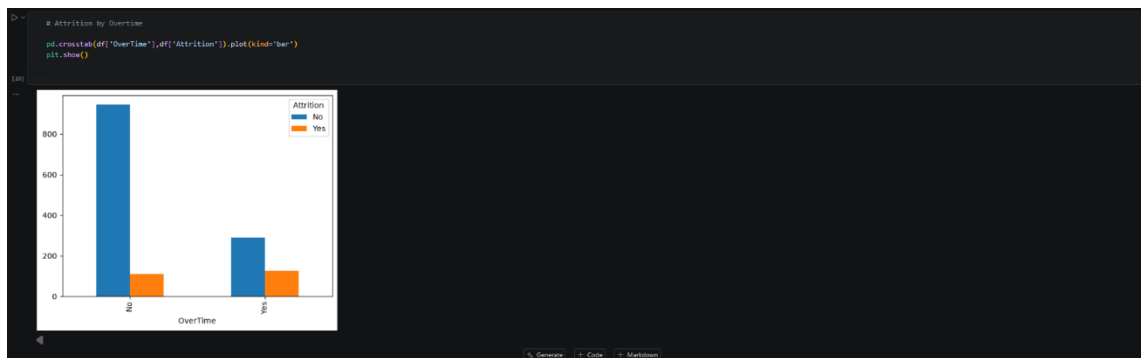
6. Attrition by Department



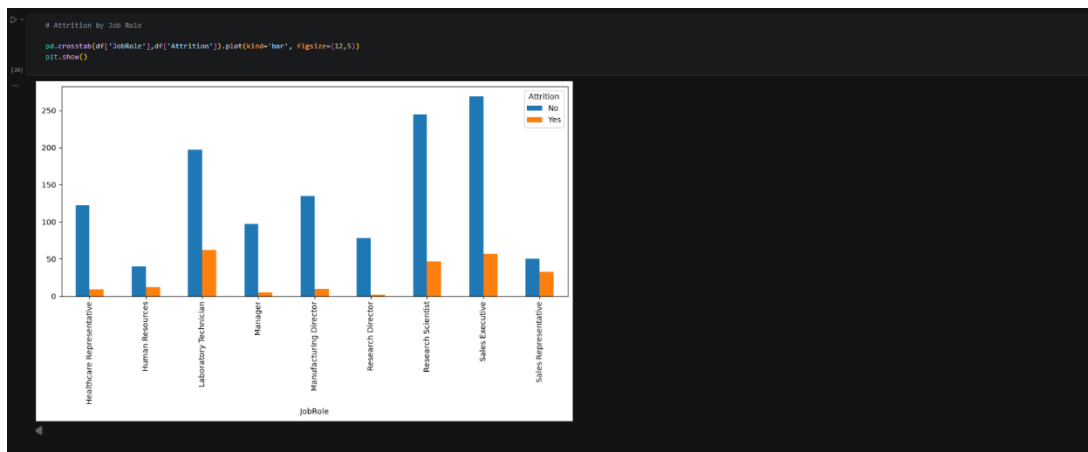
7. Attrition by Gender



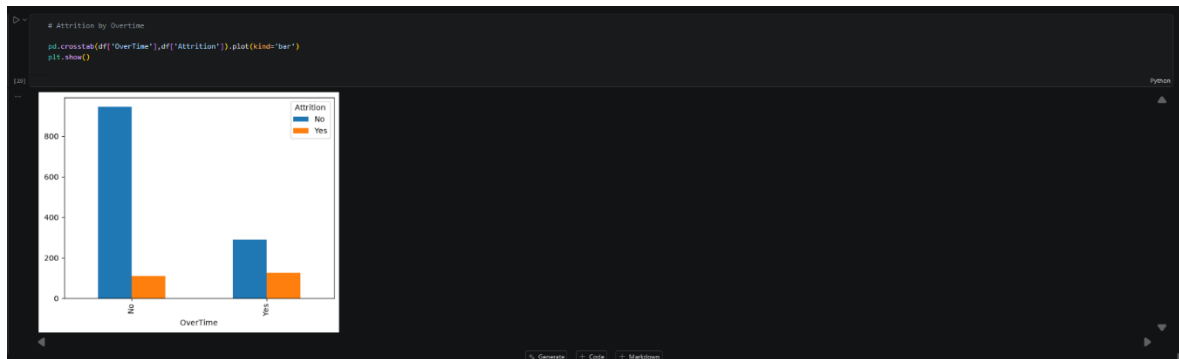
8. Attrition by overtime



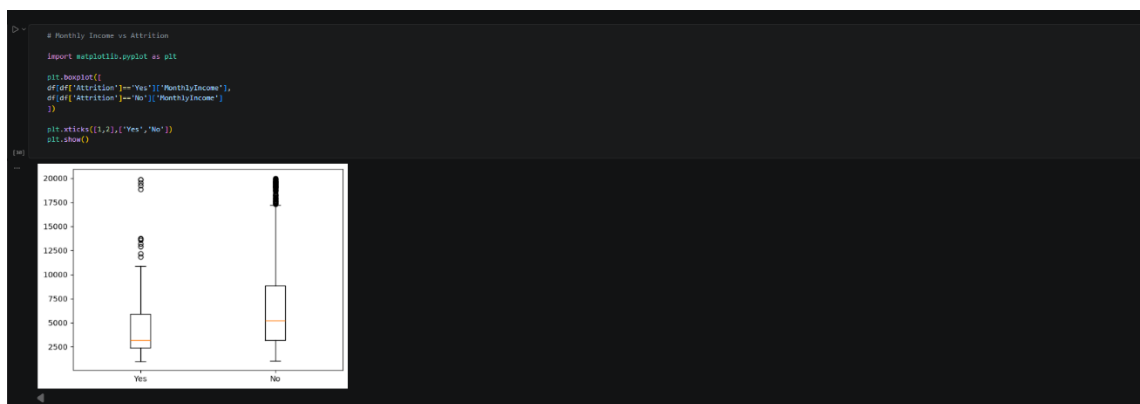
9. Attrition by Job Role



10. Attrition by Overtime



11. Monthly Income vs Attrition



Key Performance Indicator (KPI's)

```
Key Performance Indicator
markdown

# Total Employees
len(df)
[13]
-- np.int64(237)

# Attrition Count
(df['Attrition'] == "Yes").sum()
[14]
-- np.int64(237)

# Attrition Rate
(df['Attrition'] == "Yes").mean() * 100
[15]
-- np.float64(16.112448979591837)

# Average Age
df['Age'].mean()
[16]
-- np.float64(36.923889523889524)

# Average Monthly Income
df['MonthlyIncome'].mean()
[17]
-- np.float64(6502.931292517807)
```

```
# Average Years at Company
df['YearsAtCompany'].mean()
[17]
-- np.float64(7.8881632653861225)

# Average Job Satisfaction
df['JobSatisfaction'].mean()
[18]
-- np.float64(2.7285714285714286)

Exporting Dataset
markdown

df.to_csv("C:\\Users\\ucdga\\Desktop\\Internship Project\\Project Report\\cleaned_employee_attrition_python.csv", index=False)
[19]
-- Python
```

Exporting Dataset:

```
Exporting Dataset
markdown

df.to_csv("C:\\Users\\ucdga\\Desktop\\Internship Project\\Project Report\\cleaned_employee_attrition_python.csv", index=False)
[19]
-- Python
```

Data Integration and Dashboard Development

AFTER COMPLETING THE DATA CLEANING AND PREPROCESSING IN PYTHON USING THE PANDAS LIBRARY, THE CLEANED EMPLOYEE ATTRITION DATASET WAS EXPORTED AND UPLOADED INTO A MYSQL WORKBENCH DATABASE. THE MYSQL SERVER SERVED AS THE CENTRALIZED DATA SOURCE FOR THE PROJECT.

POWER BI DESKTOP WAS THEN CONNECTED TO THE MYSQL DATABASE USING THE MYSQL CONNECTOR. THE DASHBOARD WAS DEVELOPED BY IMPORTING THE CLEANED DATA DIRECTLY FROM THE MYSQL SERVER, ENSURING A STRUCTURED AND DATABASE-DRIVEN WORKFLOW. THIS APPROACH DEMONSTRATES AN INDUSTRY-STANDARD ETL (EXTRACT, TRANSFORM, LOAD) PROCESS, WHERE DATA IS CLEANED IN PYTHON, STORED IN A RELATIONAL DATABASE, AND VISUALIZED USING POWER BI.

THE POWER BI DASHBOARD INCLUDES KEY HR METRICS SUCH AS TOTAL EMPLOYEES, ATTRITION COUNT, ATTRITION RATE, AVERAGE AGE, AND AVERAGE MONTHLY INCOME. INTERACTIVE VISUALIZATIONS WERE CREATED TO ANALYZE EMPLOYEE ATTRITION ACROSS VARIOUS DIMENSIONS, INCLUDING DEPARTMENT, JOB ROLE, OVERTIME, SALARY, WORK-LIFE BALANCE, JOB SATISFACTION, EDUCATION FIELD, AND YEARS AT THE COMPANY. SLICERS WERE ADDED TO ENABLE DYNAMIC FILTERING AND IMPROVE DASHBOARD INTERACTIVITY

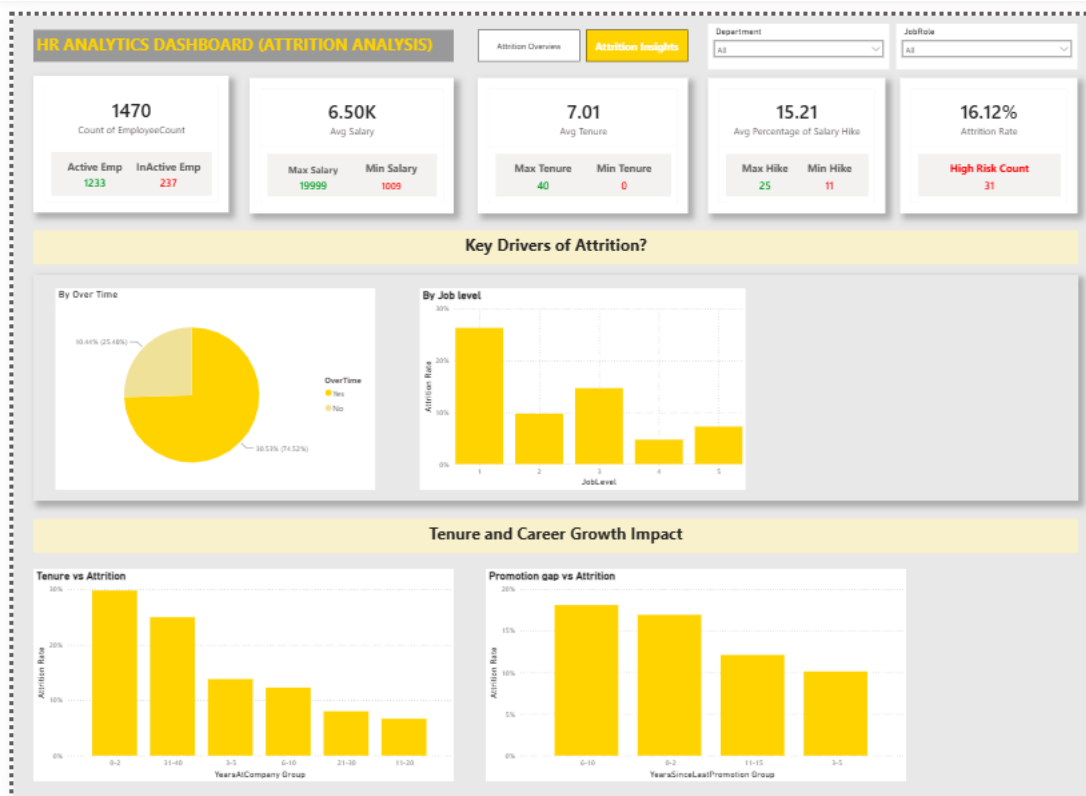


Objective

The Employee Attrition Overview Dashboard provides a comprehensive summary of the organization's workforce. It enables HR managers to monitor overall attrition trends, employee distribution, salary statistics, tenure, and demographic characteristics. The dashboard helps identify which employee groups are leaving the organization and highlights departments and job roles with higher attrition.

Key Insights

- The organization has **1,470 employees**.
- The overall employee attrition rate is **16.12%**.
- **237 employees** have left the organization.
- The Sales department experiences the highest employee attrition.
- Laboratory Technicians and Sales Executives contribute significantly to employee turnover.
- Employees between **18–25 years** exhibit the highest attrition rate.



Objective

The Employee Attrition Insights Dashboard focuses on identifying the major factors influencing employee turnover. It examines overtime, job level, employee tenure, and promotion history to understand why employees leave the organization.

Key Insights

- Employees working overtime have a higher attrition rate than employees who do not work overtime.
- Employees with lower job levels exhibit higher turnover.
- Most resignations occur within the first few years of employment.
- Employees who wait longer for promotions tend to leave more frequently.
- Early-career employees require stronger engagement and career development programs.

Business Insights :

- The overall employee attrition rate is **16.12%**, indicating a moderate level of workforce turnover.
- Sales and Human Resources departments experience comparatively higher attrition.
- Employees aged **18–25 years** are more likely to resign than other age groups.
- Overtime is strongly associated with employee attrition.
- Lower job levels experience greater employee turnover.
- Employees in the early stages of their careers show a higher likelihood of leaving the organization.
- Delayed promotions appear to contribute to increased employee attrition.

Conclusion

This project successfully analyzed employee attrition using Python, MySQL, and Power BI to identify the major factors influencing employee turnover. The raw employee dataset was cleaned and pre-processed using Python, ensuring data quality and consistency. The cleaned dataset was then stored in a MySQL database, which served as the data source for Power BI to create interactive dashboards.

Through exploratory data analysis and dashboard visualizations, several important patterns were identified. The analysis showed that employee attrition is influenced by factors such as overtime, department, job role, age group, job level, tenure, salary, and promotion history. The interactive dashboards enabled dynamic exploration of these factors and provided a comprehensive overview of workforce trends.

Based on the findings, practical business recommendations were proposed, including improving work-life balance, enhancing career development opportunities, reducing excessive overtime, strengthening employee engagement, and implementing competitive compensation and recognition programs. These recommendations can help organizations reduce employee turnover and improve workforce retention.

Overall, this project demonstrates how data analytics and business intelligence tools can transform raw HR data into meaningful insights that support strategic decision-making. It also highlights the integration of Python for data preprocessing, MySQL for database management, and Power BI for interactive reporting, providing an end-to-end analytics solution for employee attrition analysis.